Databases

Mini-project: Database design and development

Introduction

This mini-project covers all the steps of the MERISE method seen in this module and used in industry for database design and development. The project is a free choice among the 17 Sustainable Development Goals (https://sdgs.un.org/goals).

Based on a field that interests you, you will build your database by following these steps :

Step 1 : Analyze requirements

• Choose a field that interests you.

• Use the IAG to obtain information about the business rules of your field and the data to be stored.

Step 2 : CDM (Conceptual Data Model)

• Using the data obtained in step 1, produce the Entity-Relationship Model (ER Model).

• It must be normalized and include advanced modeling elements.

Step 3 : LDM (Logical Data Model) and PDM (Physical Data Model)

LDM

- Deduce the LDM from your CDM by applying the transformation rules.
- The resulting LDM must be added to the README.md documentation file.

PDM

• Deduce the PDM from your LDM in SQL : It must include integrity constraints. The whole thing must be written in a file named 1_creation.sql

- The validation constraints, which correspond to additional constraints on the data, must be listed in a file named 2_contraintes.sql . They must correspond to the business rules obtained in step 1

Step 4 : Insert data

- Populate the database using Generative Artificial Intelligence (GAI).
- Complete the provided prompt using the RICARDO framework, specifying the activity domain, context, and references without modifying its structure.
- Generate data consistent with the business rules and data dictionary.
- Ensure that the integrity constraints of the generated data are respected.
- Verify the generated data and refine the prompt if necessary.
- The result obtained must be stored in a file named 3_insertion.sql
- The text prompt and an SQL script for insertion must be included in the README.md file.

Step 5 : Query the database

- The database is ready and must now be used in a practical context.
- You need to imagine who will use this database and for what purpose. This must be related to the contextual elements you used for the design in the first step.
- Then write the corresponding queries in SQL in a file called 4_interrogation.sql . They must cover the different types of queries seen in the course :
  - **projections and selections**, with sorting, unique data, and the use of masks, IN, BET- WEEN (at least 5)
  - **aggregation functions for calculating statistics**, used with GROUP BY and HAVING, (at least 5)
  - **multiple joins** (at least 5)
  - **nested queries** with (NOT) IN, (NOT) EXISTS, ANY, ALL, etc. (at least 5)

Objectives :

The objectives to be achieved are:

- Apply current concepts: E/R model, relational model, SQL.
- Provide a satisfactory, elegant solution that respects data integrity.

Terms :

- You have 8 hours to complete this mini-project.
- The work should be done alone.

Project submission :

The project must be submitted electronically via Moodle as a single folder named with your first and last name.
The folder must contain the following files:
- README.md: Documentation including the prompt used, the AI-generated business rules and data dictionary, and a structured description of the different parts of the project.
- .loo: E/R model source file.
- 1_creation.sql: Database creation script.
- 2_contraintes.sql: Integrity constraints script.
- 3_insertion.sql: Data insertion script.
- 4_interrogation.sql: SQL queries script.
Ensure that all files are correctly named, complete, and well-organized.

Delivery date :

- The project submission date via Moodle is 26/10/2025 at 23:55 pm.